J.20

NAME ☞

SCHOOL ☞

TEACHER ☞

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# Pre-Junior Certificate Examination, 2020

# Mathematics

## Paper 2

## Higher Level

## Time: 2 hours, 30 minutes

300 marks

| For Examiner | | | |
|---|---|---|---|
| Question | Mark | Question | Mark |
| 1 | | 11 | |
| 2 | | 12 | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |
| 9 | | | |
| 10 | | Total | |

| School Stamp |
|---|
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| | |
|---|---|
| Running Total | |

| Grade |
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## Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

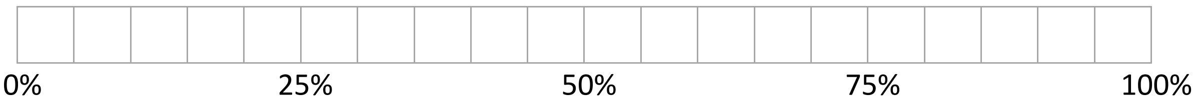
Write the make and model of your calculator(s) here:

**Question 1** **(Suggested maximum time: 5 minutes)**

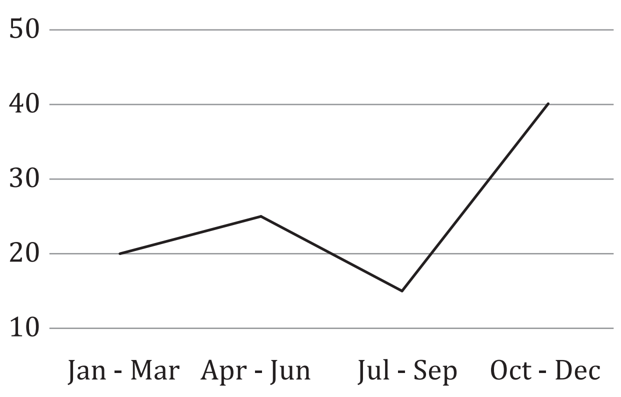
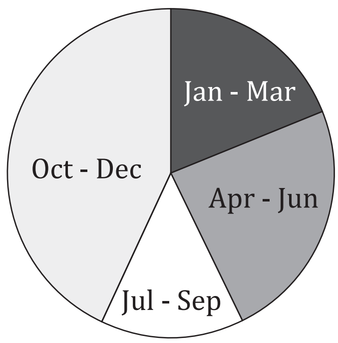
The 2019 quarterly sales figures for a company are expressed as a percentage of the total annual sales in the table shown.

| Quarter | % of Annual Sales |
|---|---|
| January - March | 20 |
| April – June | 25 |
| July – September | 15 |
| October – December | 40 |

**(a)** Complete the combined bar chart below showing the 2019 quarterly sales for the company.




These sales figures are also displayed on both the trend graph and pie chart shown below.

| 2019 Annual Sales % | 2019 Annual Sales % |
|---|---|
|  |  |





**(b)** Which of the three charts displayed above, the combined bar chart, the trend graph or the pie chart, is the best chart for displaying this data? Give a reason for your answer.

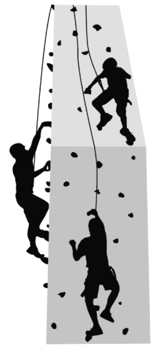
Answer:

Reason:

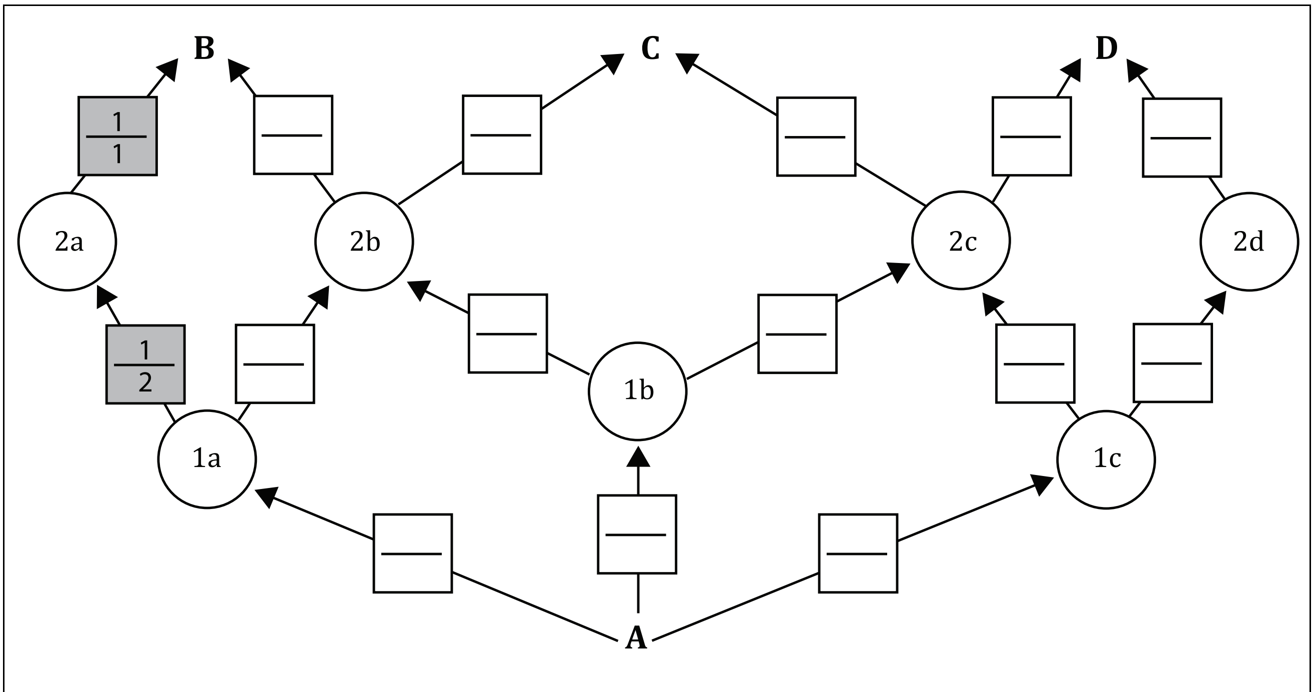
**(c)** The total sales for October to December are €12,000,000.
Find the total annual sales for the company.

**Question 2** **(Suggested maximum time: 15 minutes)**

A climbing wall is represented below. The arrows show the different routes a climber can take to the reach the top of the wall.
Climbers are equally likely to choose any of the routes available above them.
Starting at point $A$, a climber has three options to get to level 1 and so on until they finish their climb at points $B$, $C$ or $D$.



**(a)** Fill in the boxes to complete the **tree diagram** to show the probability that a climber may choose a given route from one point to another, by putting the correct fraction in the spaces provided. Two have already been filled in.




**(b)** A climber starts at point $A$, climbs higher on every move, and finishes at point $B$.
How many possible (different) routes can the climber take?

Answer = [ ] routes



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**(c)** A climber, climbing higher on every move, successfully completes the climb up to the top of the wall. Find the probability that the climber finishes at point $B$.

**(d)** Climbers are more likely to choose one of two routes to the top of the wall.
One of the routes is as follows:

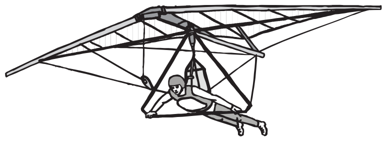
| | | | | | | |
|---|---|---|---|---|---|---|
| **A** | → | **1a** | → | **2a** | → | **B** |

Fill in the missing details below to indicate the second route?

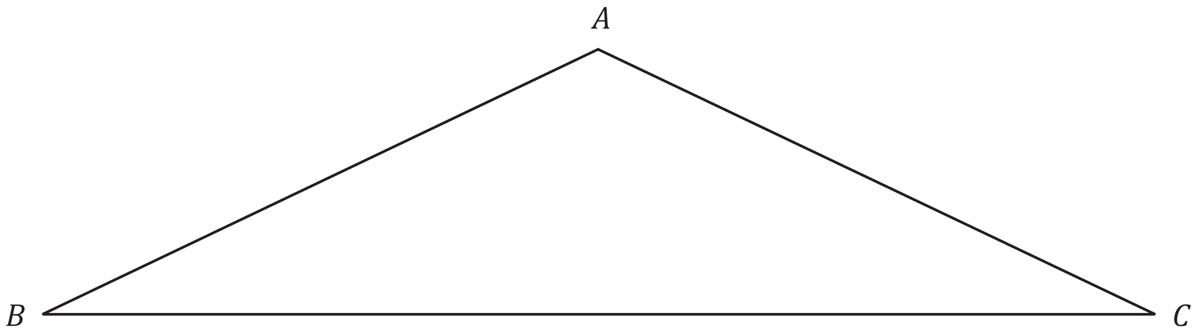
| | | | | | | |
|---|---|---|---|---|---|---|
| **A** | → | | → | | → | |

**Question 3** **(Suggested maximum time: 15 minutes)**

A hang glider is made up of a frame, cables and a wing.
The material for the wing is usually made from a synthetic sailcloth in the shape of an isosceles triangle.



A scaled diagram of a wing $ABC$ can be seen below.




**(a)** Construct the line of symmetry on the scaled diagram of the wing $ABC$ above.
You may only use a compass and straight edge. Show all of your construction lines clearly.

**(b)** Given that $|AB| = |AC| = 9{\cdot}2$ m and $|BC| = 16{\cdot}9$ m.
Work out the scale of the diagram above.
Give your answer in the form $1: n$, where $n \in \mathbb{N}$.

1 :

**(c)** Use the theorem of Pythagoras or otherwise to find the **actual length** of the line of symmetry of the wing. Give your answer in metres, correct to two decimal places.

**(d)** Use trigonometric ratios to find $|\angle ABC|$.
Give your answer in degrees, correct to one decimal place.

Answer:



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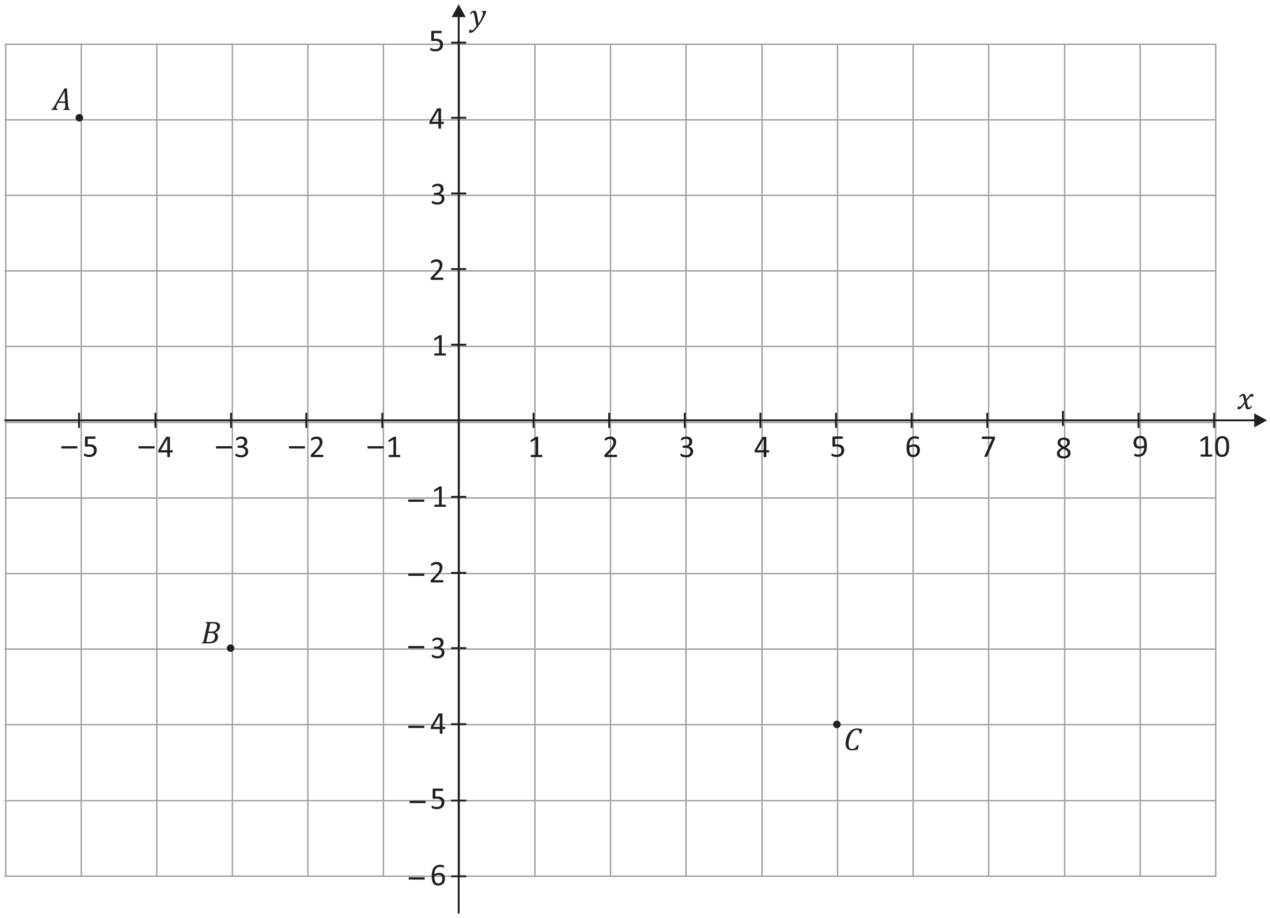
**Question 4** **(Suggested maximum time: 25 minutes)**

The points $A(-5, 4)$, $B(-3, -3)$ and $C(5, -4)$ are shown on the co-ordinate diagram below.

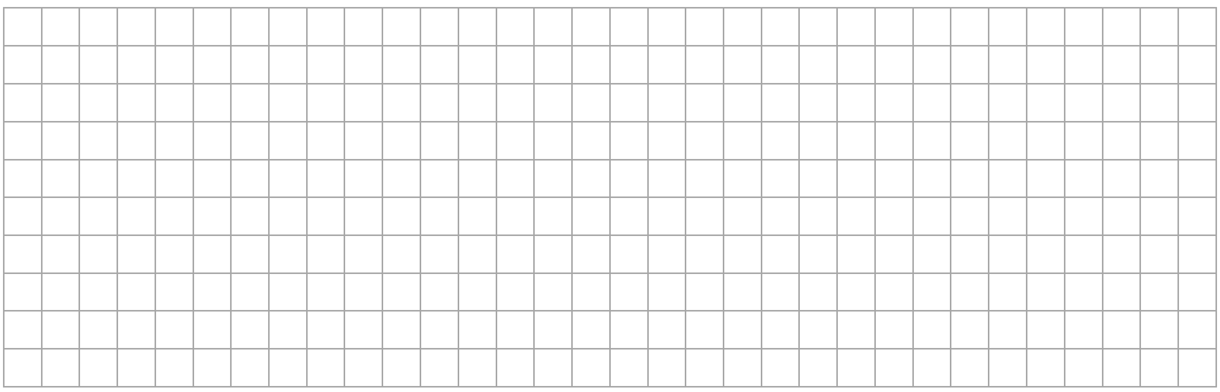



The point $D$ is the image of point $B$ under central symmetry through the origin.

**(a)** Plot the point $D$ on the co-ordinate diagram and write down it's co-ordinates below.

$D($ , $)$

**(b)** Find $|AB|$, giving your answer in surd form.



**(c)** Show that the line $BC$ is parallel to the line $AD$.

The triangles $ABC$ and $ACD$ form the parallelogram $ABCD$.

**(d)** Prove that the triangles $ABC$ and $ACD$ are congruent by completing the following table of statements and reasons.

| Statement | Reason |
|---|---|
| $\lvert AB \rvert = \lvert CD \rvert$ | |
| $\lvert AD \rvert = \lvert BC \rvert$ | |
| | Common Side |
| Triangles $ABC$ and $ADC$ are congruent | |

**(e)** Show that $4x + 5y = 0$ is the equation of the line $AC$.

*This question continues on the next page.*



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**(f)** The line $k$ is perpendicular to $AC$ and passes through the point $(5, -4)$.
Find the **equation** of the line $k$.
Give your answer in the form $ax + by + c = 0$ where $a$, $b$ and $c \in \mathbb{Z}$.

**(g)** Draw the line $k$ on the coordinate grid on page 8.



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**Question 5** **(Suggested maximum time: 10 minutes)**

The results of a test can be seen in the stem and leaf diagram below.
One of the results is missing and replaced with the letter $x$, where $x \in \mathbb{N}$, $1 \leq x \leq 50$.

| | | | | | |
|---|---|---|---|---|---|
| 2 | 0 | 2 | 7 | 9 | |
| 3 | 1 | 4 | $x$ | 8 | 8 |
| 4 | 0 | 2 | 3 | 5 | |
| 5 | 2 | | | | |

Key: | 3 | 1 = 31 marks

**(a)** The **median** test result is 37 marks. Find the value of $x$.

**(b)** Work out the **mean** of the data.

**(c)** What percentage of pupils scored more than the **modal** result?
Give your answer correct to one decimal place.

**(d)** Work out the **inter-quartile range** of the data.



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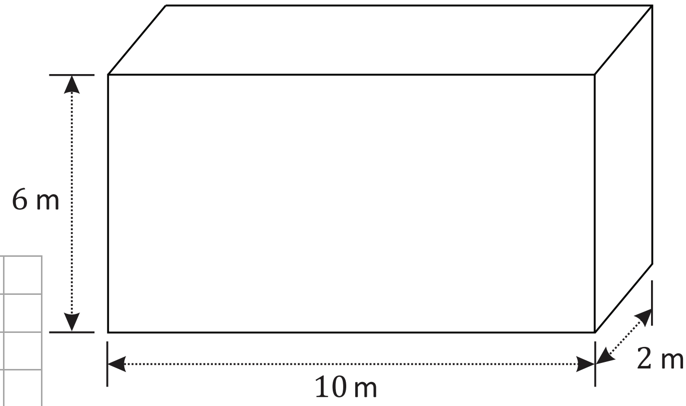
## Question 6

**(Suggested maximum time: 15 minutes)**

The cuboid shown in the diagram has dimensions, 10 m long, 2 m wide and 6 m high, as shown.



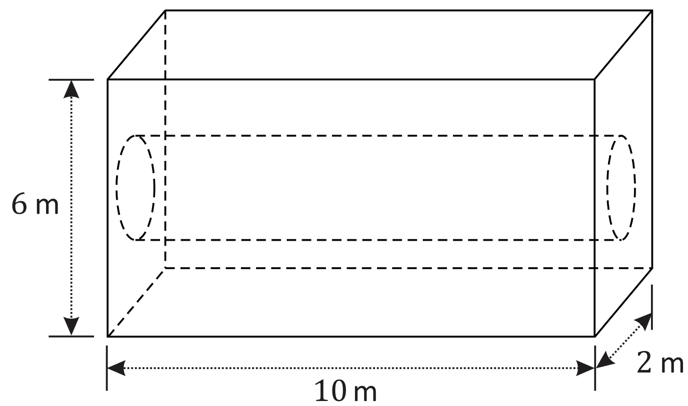

**(a)** Work out the **total surface area** of the cuboid in $m^2$.

**(b)** Draw a **net** of the cuboid using the scale 1: 100.
Show all your construction lines clearly.

**(c)** Work out the **volume** of the cuboid in $m^3$.

**(d)** A cylindrical hole has been drilled through the cuboid as shown.
The volume of the cylindrical hole is 20% of the volume of the cuboid before the hole was drilled.

Find the radius of the cylindrical hole, giving your answer correct to one decimal place.



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**Question 7** **(Suggested maximum time: 15 minutes)**

A school has a total staff of 82. It has a student population of 824.
All staff and students were asked to complete an online survey on the topic of changing the school timetable from class periods of 40 minutes to 1 hour lessons.
The results can be seen in the sample space below.

| | Time per lesson | |
|---|---|---|
| | 40 minutes | 1 hour |
| Number of staff votes | 12 | 66 |
| Number of student votes | 48 | 30 |

**(a)** What percentage of the student population took part in the survey?
Give your answer correct to one decimal place.

**(b)** Do you think the survey was a fair reflection of the opinions of the school community regarding the changing of the school timetable? Give a reason for your answer.

Answer:

Reason:

**(c)** State an advantage and a disadvantage of an online survey.

Advantage:

Disadvantage:

**(d)** Display the data graphically in a way that allows you to compare the data for students and staff. Label your graph(s) clearly.

A person is chosen from those who completed the survey.

**(e)** What is the probability that the person chosen was a student who voted for 40 minute lessons?

**(f)** What is the probability that the person chosen voted for 1 hour lessons?



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**Question 8** **(Suggested maximum time: 10 minutes)**

Prove that the angle at the centre of a circle standing on a given arc is twice the angle at any point of the circle standing on the same arc.

*Diagram*:

*Given*:

*To Prove*:

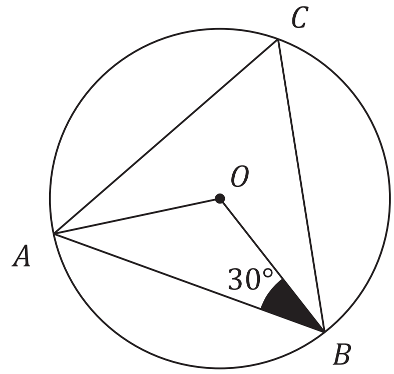
*Construction*:

*Proof*:

## Question 9

**(Suggested maximum time: 10 minutes)**

$A$, $B$ and $C$ are points on a circle as shown.
$O$ is the centre of the circle.
$|\angle ABO| = 30^\circ$.




**(a)** Find $|\angle AOB|$.

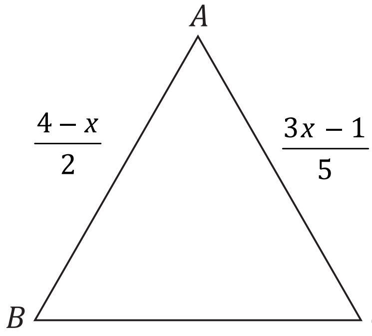
**(b)** Find $|\angle BCA|$.

**(c)** The radius of the circle is $5$ cm. Use trigonometry to find $|AB|$.
Give your answer correct to one decimal place.

**Question 10** **(Suggested maximum time: 5 minutes)**

$ABC$ is an equilateral triangle where

$|AB| = \dfrac{4-x}{2}$ m and $|AC| = \dfrac{3x-1}{5}$ m.




**(a)** Work out the value of $x$.

**(b)** Hence, work out the length of the perimeter of the triangle $ABC$.

## Question 11 **(Suggested maximum time: 5 minutes)**

**(a)** Sketch a right angled triangle were $\tan A = \frac{3}{5}$.

Label the sides that are $3$ and $5$ units in length, the right angle and the angle $A$.

**(b)** Find the measure of the angle $A$.
Give your answer in degrees, correct to the nearest whole number.

**(c)** Use your diagram, or otherwise, to show that $\sin A = \cos(90° - A)$.



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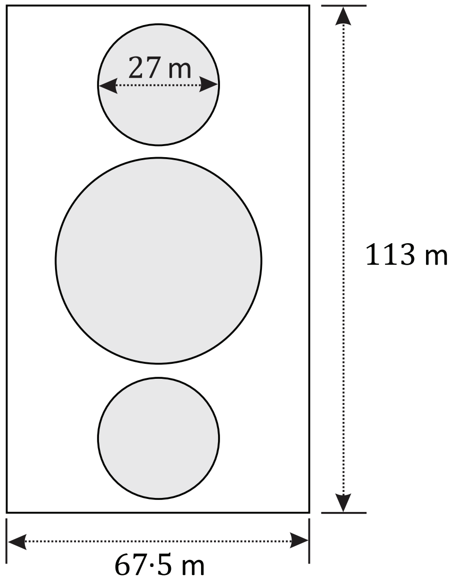
**Question 12** **(Suggested maximum time: 10 minutes)**

Crop circles are created by flattening a crop, usually cereal, growing in a field.
They became popular as UFO hoaxes in the 1980's but are now seen as modern art.

Three crop circles in a rectangular field, of length of 113 m and width of 67·5 m, are shown in the diagram below.

The two smaller crop circles are identical with a larger crop circle inbetween as shown.
The smaller circles each have a diameter of 27 m.




**(a)** Work out the **area** of one of the smaller circles.
Give your answer in terms of $\pi$.

**(b)** The larger crop circle has an area that is three times bigger than one of the smaller circles.
Work out the radius of this circle. Give your answer in the form $\dfrac{3^p}{2}$ m, where $p \in \mathbb{Q}$.

**(c)** What percentage of the field is covered by the three crop circles?
Give your answer correct to one decimal place.



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Page for extra work.
Label any extra work clearly with the question number and part.

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Label any extra work clearly with the question number and part.

Pre-Junior Certificate Examination, 2020

# Mathematics – Paper 2

Higher Level

Time: 2 hours, 30 minutes

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